



视觉与人工智能处理应用工作坊

Vision and Artificial Intelligence Processing Application

主办方
东姑阿都拉曼管理及工艺大学
ADVANCE RETRO SOLUTION SDN BHD (ARS)

宁夏职业技术大学
NINGXIA VOCATIONAL AND TECHNICAL UNIVERSITY



宁夏开放大学
NINGXIA OPEN UNIVERSITY

参与方

Vision and Artificial Intelligence Processing Application

Teaching Machines to See

DAY 1
第一天



培训团队 Our Team



张贻期 教授

研究领域 Area of expertise

计算机工程、信号处理、视频编码、嵌入式系统(物联网)、安全(信息隐藏)

学术背景 Academic background

PhD (UM), MInfTech (UKM), BEng (MMU), P. Tech, SMIEEE, AFAAET

参与活动 Activities

- 担任东南亚世界技能大赛(2018年, 2023年与2025年)物联网领域专家。
- 担任马来西亚技能大赛物联网领域专家。



培训团队 Our Team



侯漾恩 讲师

研究领域 Area of expertise

图像处理、视频处理、信号处理、视频编码、安全(信息隐藏)

学术背景 Academic background

MIT (TAR UMT), BSoftEng (Hons) (TAR UMT)

参与活动 Activities

- 担任东南亚世界技能大赛2025年物联网领域教练。
- 马来西亚技能大赛物联网领域教练。



黄伟忠 讲师

研究领域 Area of expertise

机器学习、深度学习和计算机视觉

学术背景 Academic background

BSoftEng (Hons) (TAR UMT)

参与活动 Activities

- 担任东南亚世界技能大赛2025年物联网领域教练。
- 马来西亚技能大赛物联网领域教练。

PART I

Workshop Overview

研讨会概况

欢迎加入我们,开启为期2天的沉浸式实战之旅!共同探索计算机视觉、人工智能处理与机器学习的技术

Day 1 Itinerary (22 May 2026)

第一天日程安排

Time / 时间	Itinerary / 行程
09:00 AM	Welcoming & TAR UMT Gallery Visit
10:00 AM	Workshop Part 1: AI Vision Fundamentals
01:00 PM	Lunch Break @ FM Cafe
02:00 PM	Online Examination (Ningxia/ARS)
04:00 PM	Workshop Part 2: AI Vision Fundamentals

Day 2 Itinerary (23 May 2026)

第二天日程安排

Time / 时间	Itinerary / 行程
09:00 AM	Recap on Day 1 Lesson
09:30 AM	Workshop Part 3: Cloud AI and Machine Learning
01:00 PM	Lunch Break (Packed Food)
02:00 PM	Workshop Part 4: Image Processing and Video Processing
05:00 PM	Closing Ceremony

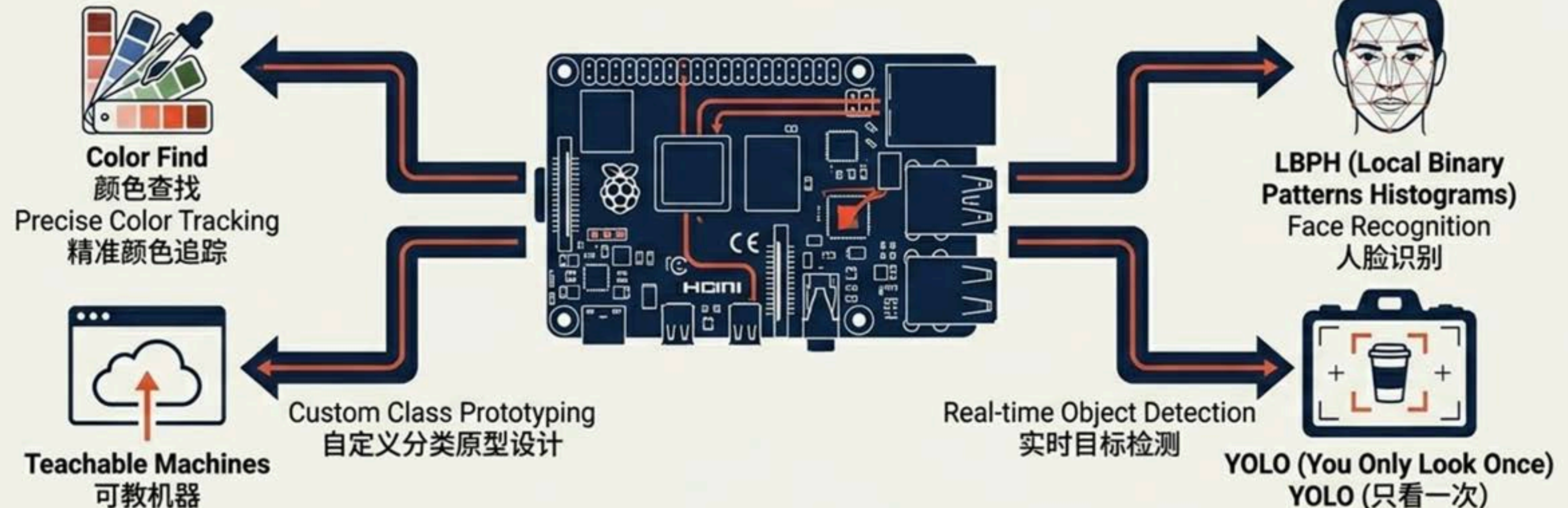
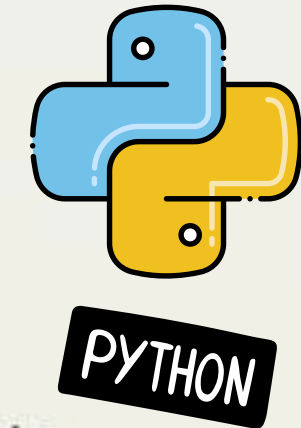


The Edge AI Ecosystem

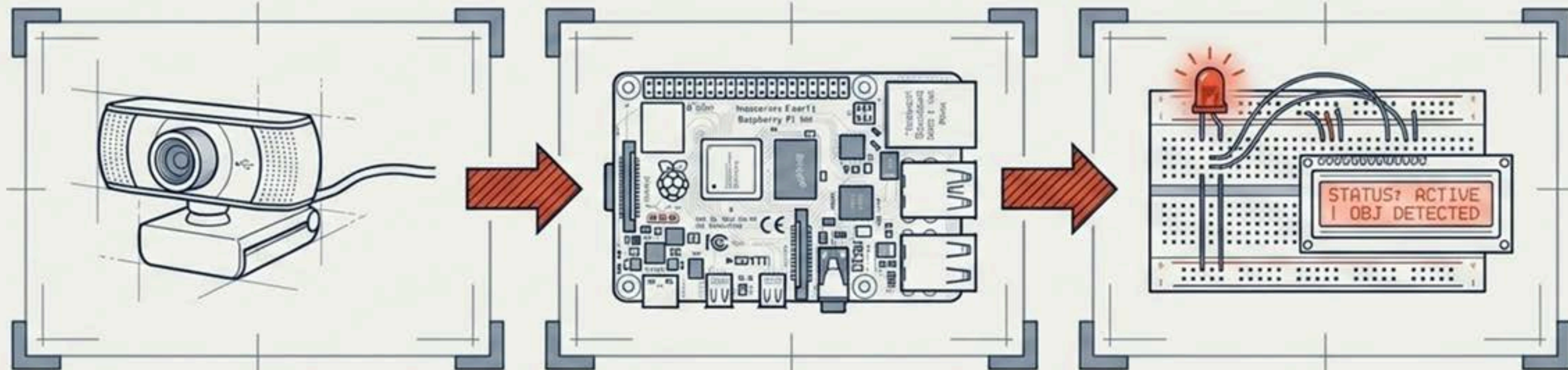
边缘人工智能生态系统

One piece of hardware. Infinite applications. The **Raspberry Pi** routes raw visual data to the appropriate processing algorithm depending on the task.

一个硬件平台，无限应用可能。树莓派根据任务的不同，将原始视觉数据路由到相应的处理算法。



How a Computer Sees 计算机如何看世界



The Digital Eye (数字之眼)
Capturing **raw pixel data** via
USB Camera.

The Edge Brain (边缘大脑)
Analyzing frames locally on the
Raspberry Pi using **Python**.

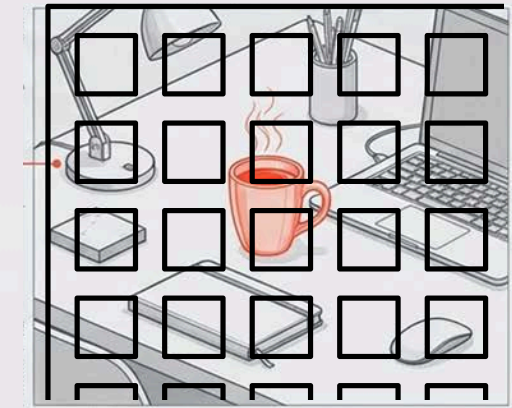
The Physical Action (物理响应)
Translating digital recognition into
real-world signals (**GPIO, LED, LCD**).

Layer 1: Color Recognition

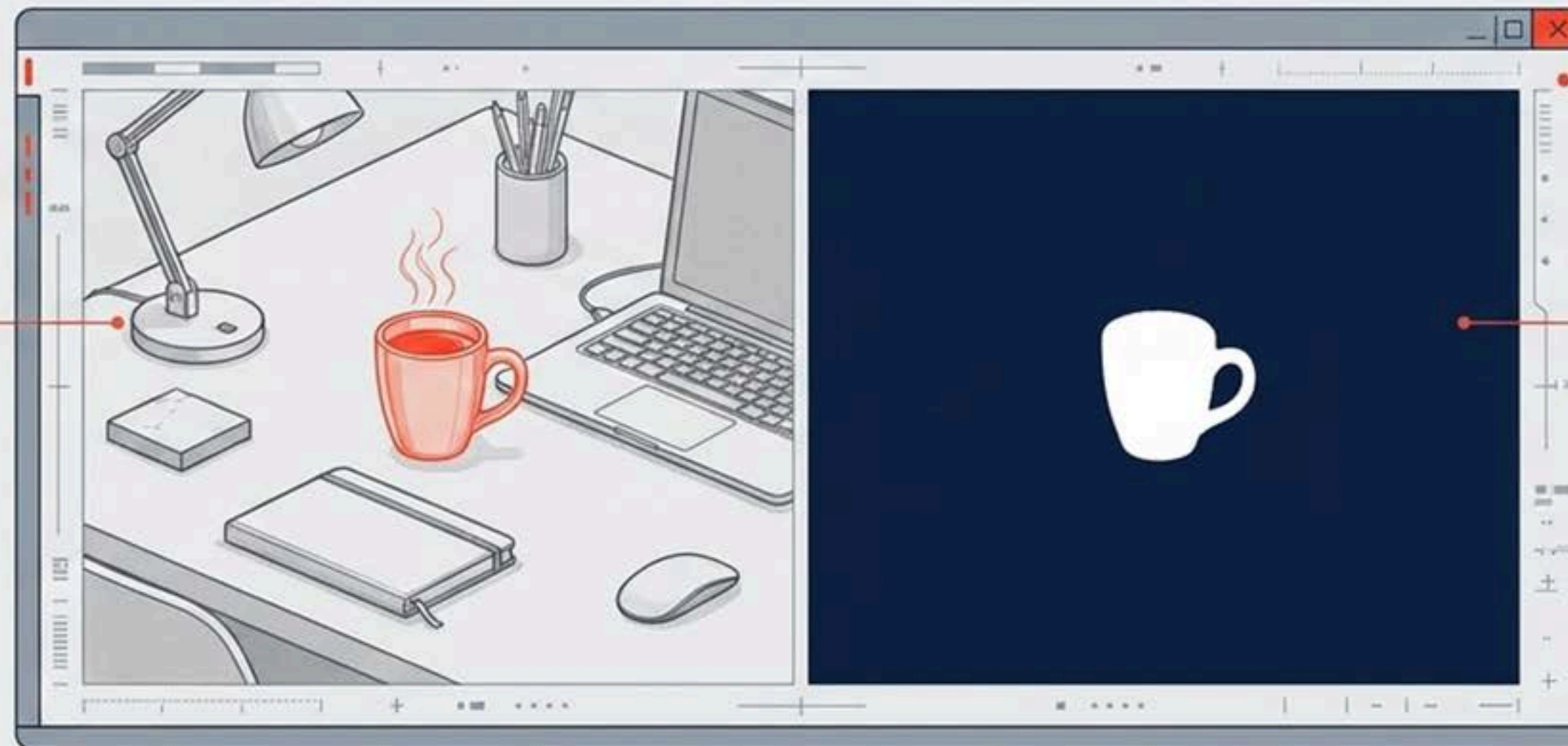
第一层：颜色识别

Before a machine understands objects, it understands pixel values. Using OpenCV, we isolate specific colors by filtering their digital signature.

在机器理解物体之前，它首先理解像素值。使用 OpenCV，我们通过过滤特定的数字特征来分离颜色。



OpenCV (cv2): Filters the image matrix to find specific color boundaries.
(过滤图像矩阵以寻找特定颜色边界)



PyQt5 GUI: Creates the visual dashboard for human-computer interaction.
(为构建人机交互界面)

Layer 2: Facial Detection (Haar Cascades)

第二层：人脸检测

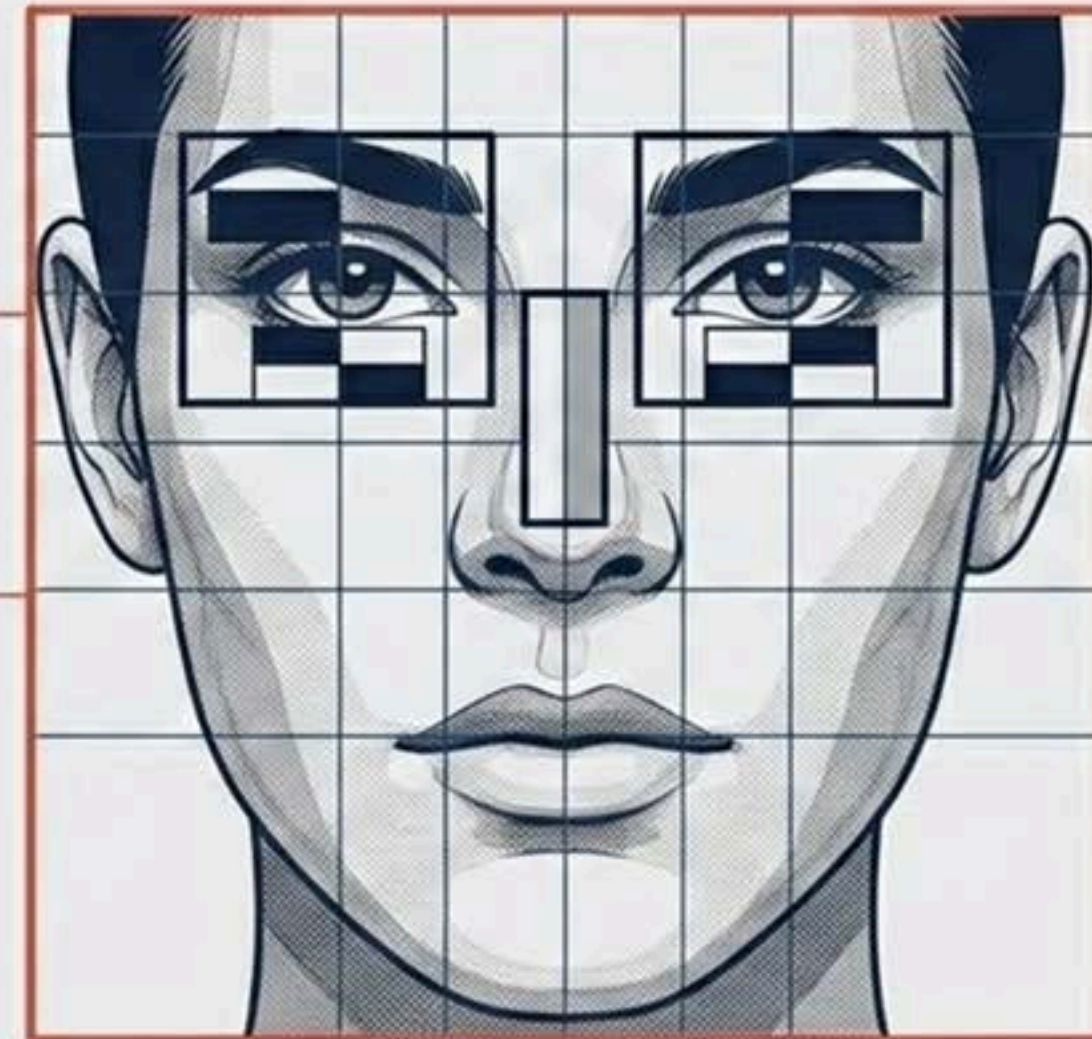
Detection is not recognition. The machine simply finds a pattern of light and dark contrast that matches a structural map of a face.

检测不等于识别。机器只是在寻找与人脸结构图相匹配的明暗对比模式。

Key Tech: `haarcascade_frontalface_default.xml`

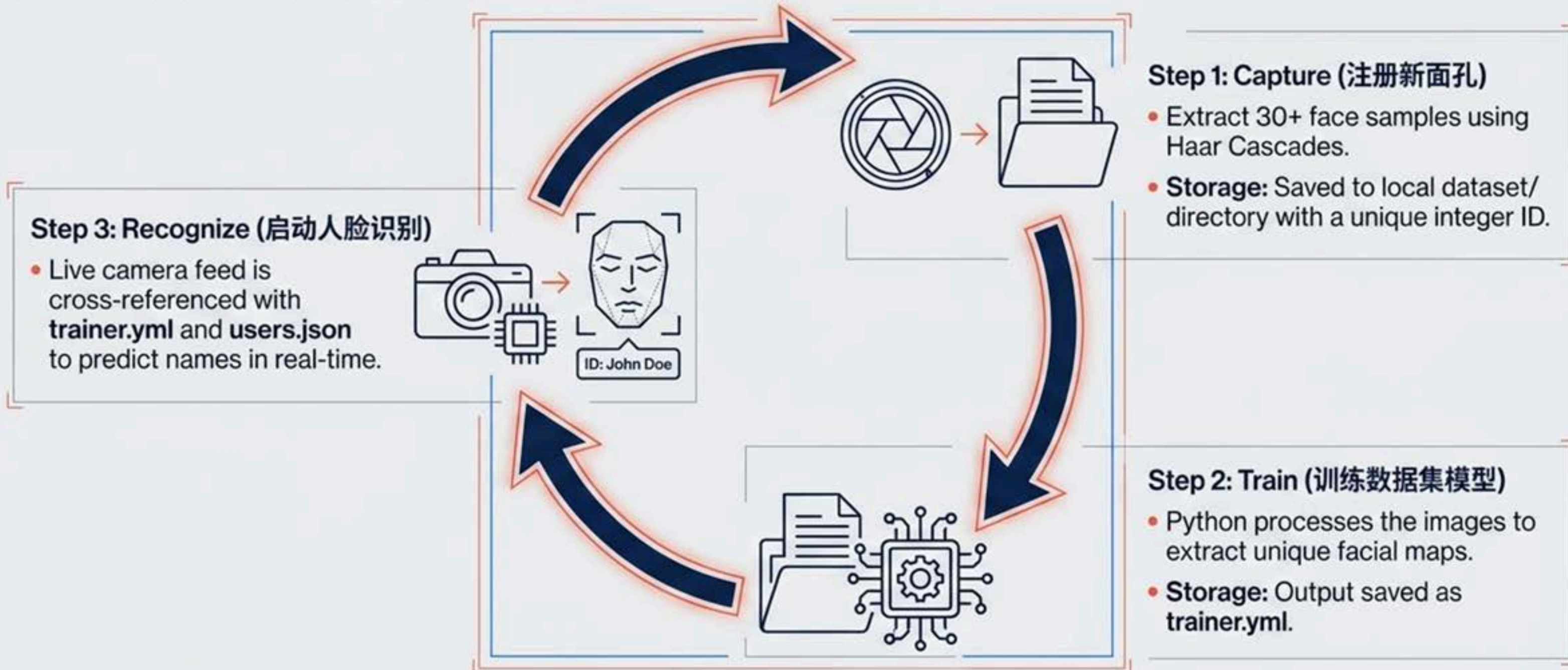
Sliding Window (滑动窗口)

- The algorithm scans the camera feed block by block, analyzing contrast.



The Face Recognition Lifecycle

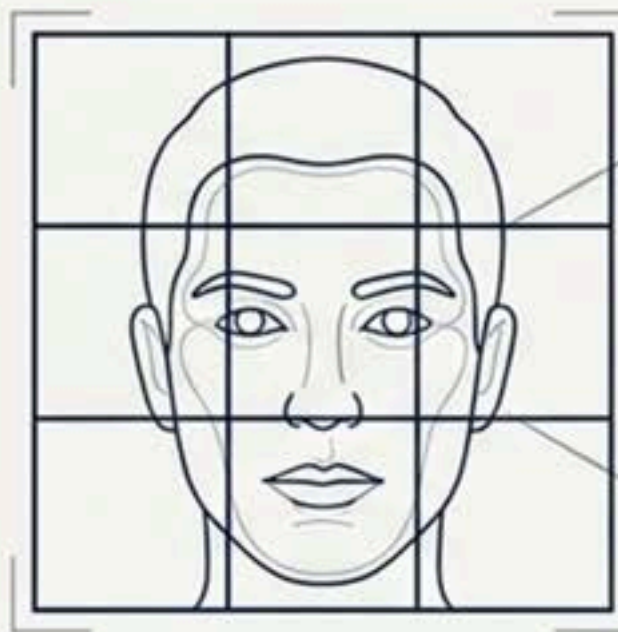
人脸识别生命周期



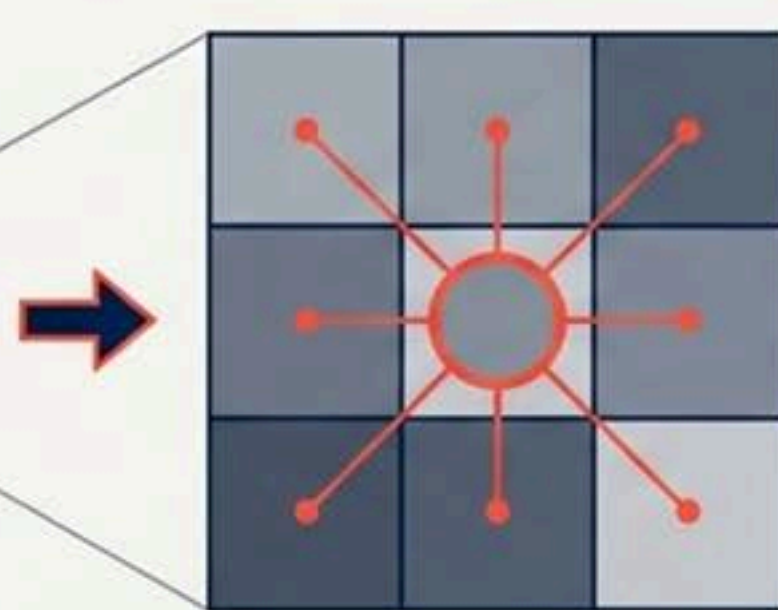
How the Machine Learns: LBPH

机器如何学习：局部二值模式直方图（LBPH）

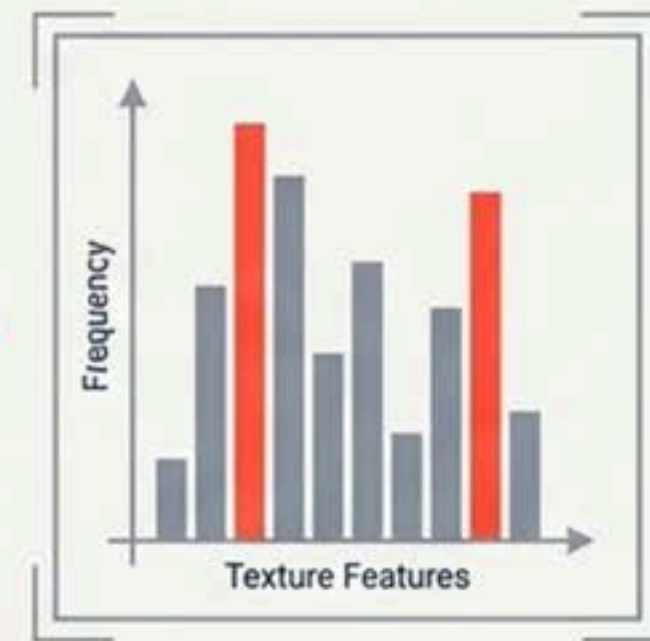
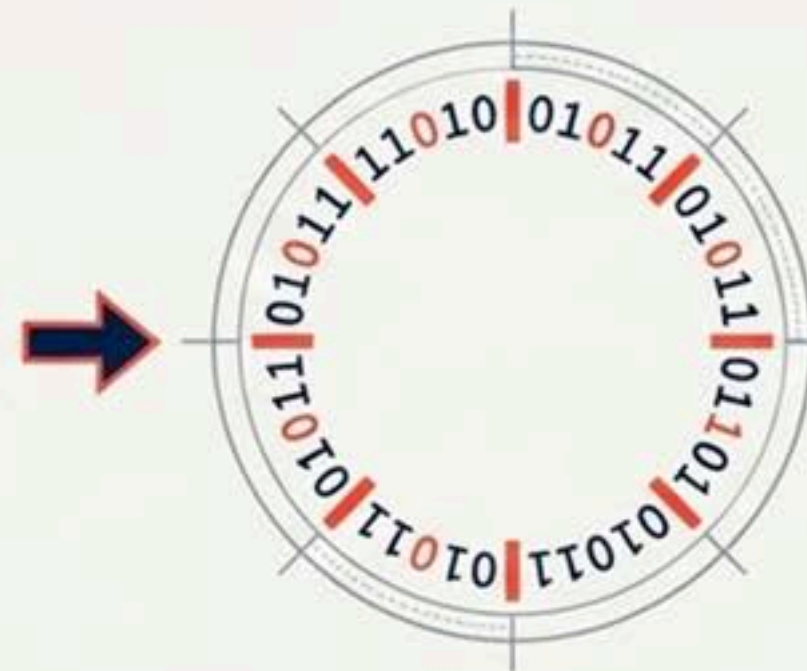
The Pi doesn't memorize a photograph.
It extracts a mathematical histogram of
micro-textures across your face.
树莓派并不记忆照片。它提取的是你脸部
微观纹理的数学直方图。



Divide: Split the face into a grid. (将人脸划分为网格)



Compare: Compare a center pixel to its neighbors. (将中心像素与其相邻像素进行比较)

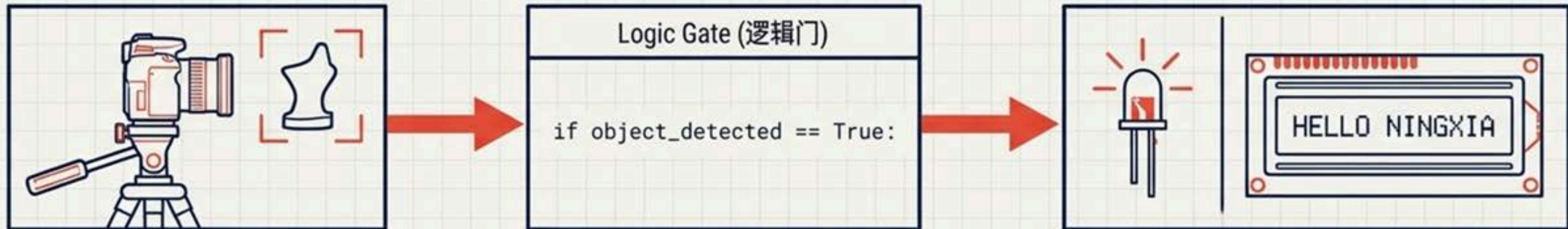


Histogram: Compile the results into a unique digital signature. (将结果汇编成独特的数字特征)

Afternoon Mission: Hardware Bridging

下午任务：硬件桥接

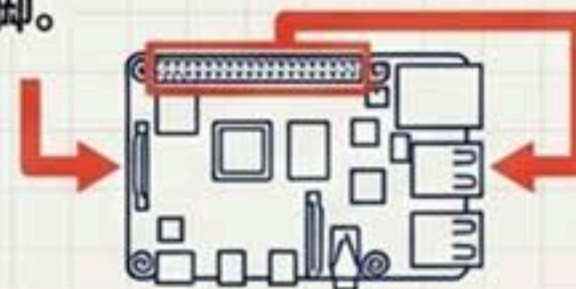
The Goal (目标): Trigger physical world outputs using digital world detection. (利用数字世界的检测触发物理世界的输出。)



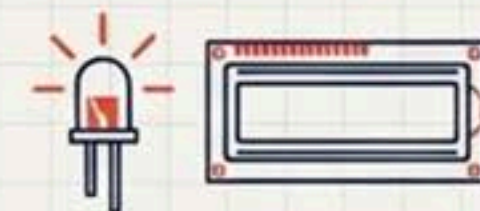
Task 1: Utilize the camera to achieve positive detection (Face or YOLO).
1. 使用摄像头实现目标检测。



Task 2: Route the Python success state to the Pi's GPIO pins.
2. 将 Python 的成功状态路由到树莓派的 GPIO 引脚。



Task 3: Successfully illuminate the LED and display confirmation text on the LCD1602.
3. 成功点亮 LED 并在 LCD1602 上显示确认文本。

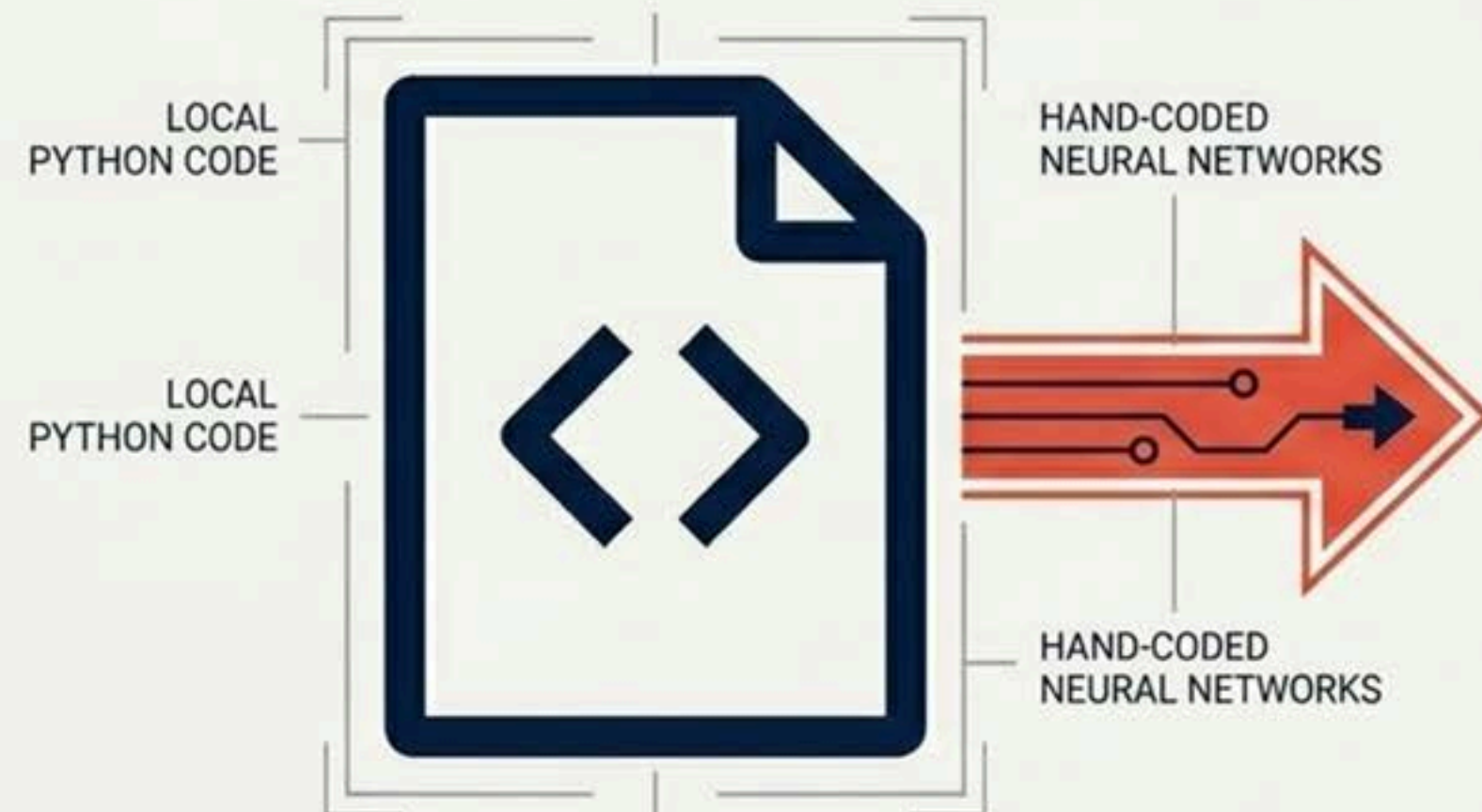


Layer 3: Teachable Machines

第三层：可教机器（Teachable Machines）

Shifting from local code to web-based neural networks. A zero-code environment to understand deep learning classification.

从本地代码转向基于网络的神经网络。
一个零代码环境，用于理解深度学习分类。



Layer 4: YOLO Object Detection

第四层：YOLO 目标检测

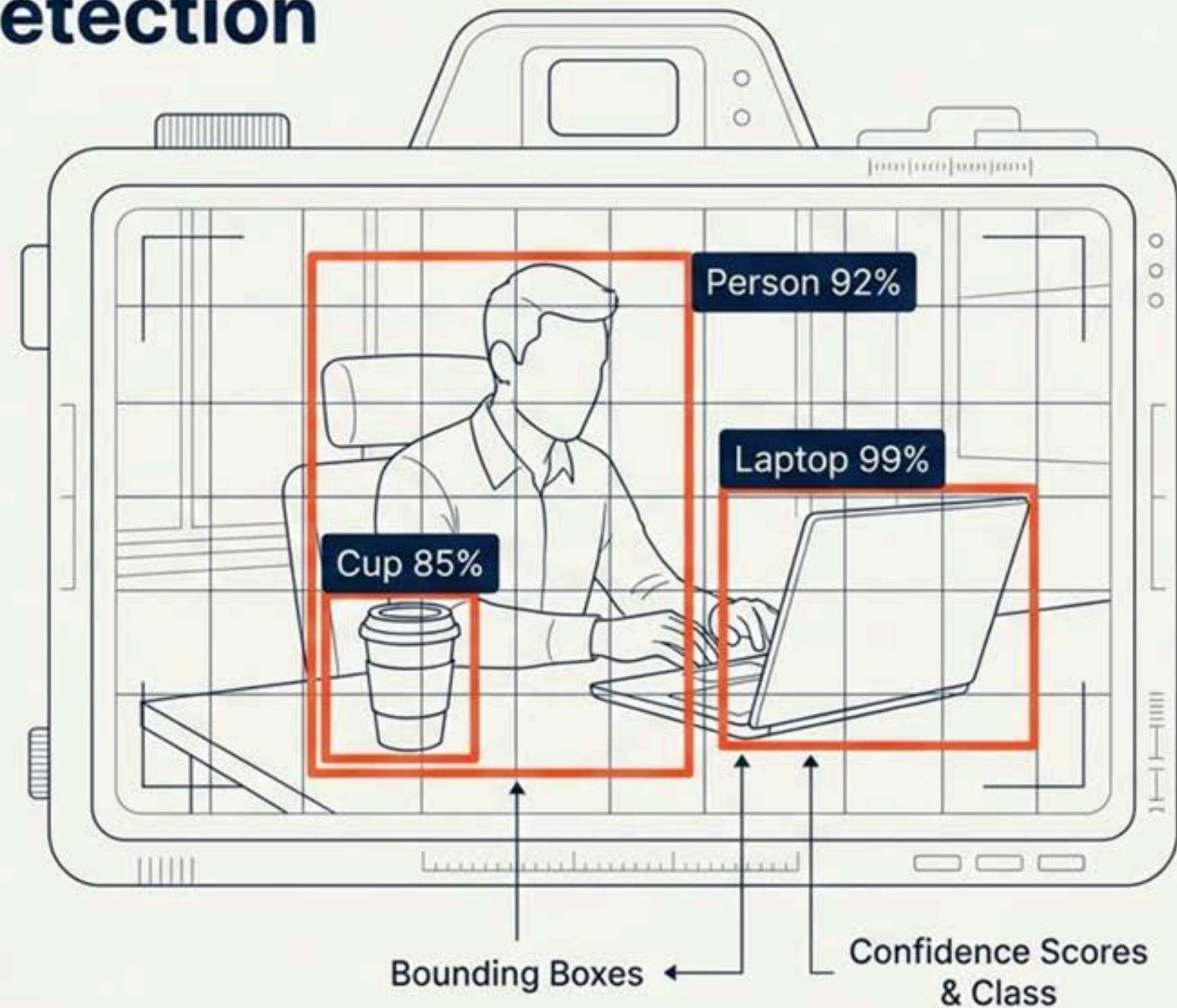
"You Only Look Once." Unlike classical computer vision that scans slowly, YOLO predicts multiple bounding boxes and class probabilities simultaneously across a single grid.

“你只看一次”。不同于缓慢扫描的经典计算机视觉，YOLO 在单个网格上同时预测多个边界框和类别概率。

✓ Local Edge Processing (本地边缘处理)

✓ Multi-class detection (多类别检测)

✓ Confidence Scores (置信度得分)



The AI Processing Matrix

人工智能处理矩阵

	OpenCV (Classical) OpenCV (经典)	Teachable Machines 可教机器	YOLO (Deep Learning) YOLO (深度学习)
Process Location (处理位置)	Local Edge (Raspberry Pi) 本地边缘 (树莓派)	Cloud Web Browser 云端网络浏览器	Local Edge AI 本地边缘人工智能
Complexity (复杂度)	Code-heavy, explicit logic 代码繁重, 显式逻辑	No-code, rapid 无代码, 快速	Pre-trained weights, advanced 预训练权重, 高级
Best Use Case (最佳应用)	Color tracking, precise geometric faces 颜色跟踪, 精确的几何人脸	Custom class prototyping 自定义分类原型设计	Real-time multi-object detection 实时多目标检测

The image features a vibrant, futuristic cityscape at night, with a digital overlay of various data visualizations. The city skyline is composed of numerous skyscrapers, including a prominent tower with a spherical observation deck on the right. The buildings are illuminated with a mix of blue and purple lights. Overlaid on the city are various digital elements: a grid of data points, line graphs, and circular patterns. A large, glowing blue sphere is positioned in the upper left, and a series of glowing lines connect different points across the scene. The overall color palette is dominated by blue and purple, creating a high-tech, digital atmosphere. The text 'Thank You!' is centered in the middle of the image, rendered in a bold, white, sans-serif font. The city and its digital elements are reflected in a shimmering, blue surface at the bottom of the frame, which appears to be a body of water or a highly reflective floor. The reflection is slightly blurred, adding to the sense of depth and movement. The text 'Thank You!' is the central focus, conveying a message of gratitude against this complex, futuristic backdrop.

Thank You!



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